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Lab 07 Model evaluation
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Reflection on Better Model Evaluation

In this lab, our group explored different ways to evaluate machine learning models beyond simply checking their accuracy. We continued working with the Titanic dataset and used Logistic Regression to predict whether passengers survived. However, instead of focusing only on overall accuracy, we examined the Confusion Matrix, Classification Report, and Cross-Validation scores. This lab connected directly to our previous work with Regression and Classification Models, where we learned how to train models and evaluate their basic performance. In Lab 06, we wondered whether accuracy alone could tell us which types of predictions our model struggled with. This time, we had an opportunity to explore that question and understand why different evaluation metrics are important.

The Confusion Matrix helped us understand how a model's predictions can be correct or incorrect in different ways (Tien). We used the confusion_matrix function to compare actual passenger survival with the model's predictions and created a heatmap to visualize the results (Tien). Our model correctly predicted 90 passengers who did not survive and 54 passengers who survived (Tien). However, it also incorrectly predicted that 15 passengers survived when they did not, and that 20 passengers died when they actually survived (Tien). Looking at these numbers helped us understand the difference between False Positives and False Negatives (Tien). One thing that required careful attention was identifying which values represented actual outcomes and which represented predictions (Tien). We realized that even when a model achieves good accuracy, it can still make mistakes that matter depending on the situation (Tien).

The Classification Report gave us a more detailed understanding of the model's performance through Precision, Recall, and F1-Score (Viktoriya). Our model reached about 80% accuracy, but when we looked at the individual scores, we noticed that it performed better for one class than the other (Viktoriya). For passengers who survived, the Precision was 0.78, Recall was 0.73, and F1-Score was 0.76 (Viktoriya). Meanwhile, the model achieved higher scores when identifying passengers who did not survive (Viktoriya). This difference was interesting because the overall accuracy initially made the model seem equally reliable for both outcomes (Viktoriya). We learned that Precision measures how often positive predictions are correct, while Recall measures how many actual positive cases the model successfully identifies (Viktoriya). The F1-Score helped us see how Precision and Recall work together to give us a better idea of the model's performance (Viktoriya). This changed how we interpret classification results since a single accuracy value does not explain every part of a model's performance (Viktoriya).

Another important part of this lab was understanding why Precision and Recall matter differently depending on the problem we want to solve (Tien and Viktoriya). The Knowledge Check encouraged us to think about real-world situations rather than only memorizing formulas (Tien and Viktoriya). For example, Precision is especially important in spam detection because

incorrectly marking an important email as spam could cause someone to miss valuable information (Tien). Meanwhile, Recall is more important in medical screening because missing a serious disease could have dangerous consequences (Viktoriya). These examples helped us understand that the best evaluation metric depends on the consequences of incorrect predictions (Tien and Viktoriya). We also started thinking about how this connects to our Titanic model and whether it would be better to focus on correctly identifying more survivors rather than just trying to improve the overall accuracy (Viktoriya).

Cross-Validation introduced another way to determine how reliable our model's results are (Zaid). We performed 5-fold Cross-Validation, which divided the dataset into five parts and evaluated the model multiple times using different training and validation sets (Zaid). Our model reached an average Cross-Validation score of about 78.68%, with a standard deviation of 0.0094 (Zaid). The individual scores were relatively close to each other, which suggested that the model performed consistently across the five folds (Zaid). This helped us understand why relying on a single train-test split can sometimes be misleading (Zaid). A model might receive a higher or lower score depending on which observations end up in the testing set (Zaid). Cross-Validation gave us a better way to examine performance across different portions of the dataset, although we also learned that consistent scores do not necessarily mean the model has no weaknesses (Zaid).

This lab also made us think more carefully about how our previous machine learning knowledge connects to model evaluation (Tien, Zaid, and Viktoriya). In Lab 06, our Logistic Regression model reached 74.83% accuracy, while in this lab, we got about 80% (Tien, Zaid, and Viktoriya). However, we noticed that the models did not use exactly the same features or data preparation methods (Viktoriya). This time, we included Fare as an additional feature and filled missing Age values using the median instead of removing those observations (Viktoriya). These differences made us wonder how much the feature selection and data preparation affected the results (Tien). We would like to compare the models under the same conditions and experiment with different classification thresholds to see whether we could improve Recall for surviving passengers without reducing Precision too much (Viktoriya). We also want to look more closely at how well a model performs across different parts of the dataset instead of judging it by just one accuracy score (Zaid).

In conclusion, this lab helped our group better understand how to evaluate machine learning models and why we should look at more than just accuracy. The Confusion Matrix showed us the different types of prediction errors, while Precision, Recall, and F1-Score explained those errors in more detail. Cross-Validation showed us why evaluating a model multiple times can provide a more reliable estimate of its performance. One of the biggest things we learned was that high accuracy does not always mean a model makes good predictions in every situation. We now understand that evaluating machine learning models requires looking at several metrics, considering the consequences of incorrect predictions, and questioning how reliable the results actually are. In future labs, we want to focus more on these details instead of just trying to get the highest accuracy score.